INTICS Model Radar

Test Case Catalogue — QA & Acceptance Reference

Hugging Face model release tracker · Spring Boot + Vaadin · Generated 2026-07-28

INTICS Model Radar — Test Cases

Comprehensive test catalogue for the Model Radar (Spring Boot + Vaadin, single app). Use it for manual QA, as an acceptance checklist, and as the backlog for automated tests.

Conventions

Type

Code	Meaning
`UNIT`	Pure unit test (no Spring, no DB, no network)
`INT`	Integration (Spring context and/or DB and/or live Hugging Face)
`API`	REST endpoint test (HTTP against the running app)
`UI`	End-to-end through the Vaadin dashboard in a browser
`OPS`	Operational / manual (scheduler, deployment, resilience)

Priority: P1 = critical (blocks "done"), P2 = high, P3 = nice-to-have.

Standing preconditions (assumed unless a case says otherwise): PostgreSQL is up (`docker compose up -d` , host `:5433`); the app is running (`:8080`); the database has been populated by at least one sync.

A. Hugging Face API client (`HuggingFaceClient`)

ID	Objective	Steps	Expected result	Type	Pri
`TC-HF-01`	Fetch by pipeline tag returns newest-first results	Call `fetchModelsByPipelineTag("image-text-to-text", 5)`	Non-empty list, ≤5 items, each with a non-blank `id` , ordered by `createdAt` desc	INT	P1
`TC-HF-02`	Fetch by keyword search returns results	Call `searchModelsByKeyword("OCR", 5)`	Non-empty list; ids/authors relate to the keyword	INT	P1
`TC-HF-03`	Request includes `cardData` + `full`	Inspect the built URL / DEBUG log	URL has `cardData=true&full=true` ; responses include `lastModified` and card license	INT	P2
`TC-HF-04`	Rate-limit (429) → retry once, then skip	Simulate/observe a 429 from HF	One retry after `retry-backoff` ; on continued 429 → empty list + WARN "rate-limited"; sync continues	INT	P2
`TC-HF-05`	Network/timeout error is swallowed	Point `base-url` at an unreachable host; run a fetch	Returns empty list, logs "fetch failed", does not throw	INT	P1
`TC-HF-06`	README fetch → short description	Call `fetchDescription(id)` for a model with a README	Returns cleaned first-paragraph text; for a model with no README (404) → `null`	INT	P2
`TC-HF-07`	Redirects on raw README are followed	Fetch a README URL that 302-redirects	Content is returned (HttpClient `followRedirect`)	INT	P3

B. Metadata extraction (`HuggingFaceMetadata`) — UNIT

ID	Objective	Input	Expected result	Type	Pri
TC-META-01	License from `cardData.license` (string)	cardData license = `"apache-2.0"`	`"apache-2.0"`	UNIT	P2
TC-META-02	License from `cardData.license` (list)	license = `["mit","other"]`	`"mit, other"`	UNIT	P3
TC-META-03	License from `license:*` tag fallback	no cardData; tags include `license:gemma`	`"gemma"`	UNIT	P2
TC-META-04	License unknown → null	no cardData license, no license tag	`null`	UNIT	P2
TC-META-05	Model size regex	ids ending `-7B` , `-1.5B` , `-70M` , `-13B` , and one with none	`7B` , `1.5B` , `70M` , `13B` , and `null`	UNIT	P2
TC-META-06	Description cleaning	README with YAML front-matter + `<a>` / ` ` badge line + first prose paragraph	Front-matter/HTML/markdown stripped; first real paragraph; ≤300 chars; empty README → `null`	UNIT	P2

C. Categorisation & tracking catalog

ID	Objective	Steps	Expected result	Type	Pri
TC-CAT-01	Pipeline tag → correct category	For each tracked tag, sync a model carrying it	`matchedCategory` matches the yml grouping (e.g. `optical-character-recognition` →OCR)	INT	P1
TC-CAT-02	Keyword → correct category	Keyword-only match (tag not tracked)	Filed under the keyword's category, `matchStrategy=KEYWORD_SEARCH`	INT	P1
TC-CAT-03	Exactly one category chosen	Model matching multiple tags	Single `matchedCategory` per the documented tie-breaks (image-to-text→OCR, image-segmentation→LAYOUT, object-detection→DIAGRAM_ANALYSIS)	INT	P2
TC-CAT-04	Only Section 4 categories tracked	Query `select distinct matched_category`	Values ⊆ {VLM,LLM,OCR,DIAGRAM_ANALYSIS,DOCUMENT_QA,LAYOUT,EMBEDDING,OTHER}	INT	P1

D. Sync & diff (`ModelSyncService`)

ID	Objective	Steps	Expected result	Type	Pri
TC-SYNC-01	First sight inserts, stamps firstSeenAt	Empty DB → run sync	New rows inserted; `first_seen_at ≈ now` ; `matchStrategy` set	INT	P1
TC-SYNC-02	Re-sync updates mutable fields only, preserves firstSeenAt	Run sync twice	Existing rows: `downloads/likes/lastModified/lastFetchedAt` refreshed; `first_seen_at` UNCHANGED	INT	P1
TC-SYNC-03	Keyword pass skips ids caught by tag pass	Run a full sync	No duplicate rows; each id processed once (`checked` = distinct ids)	INT	P1
TC-SYNC-04	matchStrategy provenance	Inspect rows	Tag-caught → `PIPELINE_TAG` ; keyword-only → `KEYWORD_SEARCH`	INT	P2
TC-SYNC-05	Description fetch + backfill, capped	Null-out descriptions; run sync	Up to `readme-fetch-limit` descriptions fetched per run; new rows get one; log shows "(N descriptions fetched)"	INT	P2
TC-SYNC-06	Summary counts correct	Run sync	`checked = new + updated` ; returned summary matches DB deltas	INT	P2

ID	Objective	Steps	Expected result	Type	Pri
TC-SYNC-07	Run recorded in audit trail	Run sync	A `sync_runs` row (RUNNING→SUCCESS) with counts	INT	P1

E. Persistence & schema

ID	Objective	Steps	Expected result	Type	Pri
TC-DB-01	Flyway migrations apply cleanly	Start app on a fresh DB	V1/V2/V3 applied; `flyway_schema_history` all success; Hibernate `validate` passes	INT	P1
TC-DB-02	All Section 5 fields stored	`\d tracked_models` + sample rows	Every field present; nullable ones null-with-reason (pipeline_tag/license/model_size/short_description)	INT	P1
TC-DB-03	Data survives restart	Restart app	Rows persist (Docker volume); counts unchanged; NEW badges age from stored timestamps	OPS	P1

F. "NEW" badge / new-since logic

ID	Objective	Steps	Expected result	Type	Pri
TC-NEW-01	Recent model is NEW	Set a row `first_seen_at = now`	Appears in `/api/models/new` ; `isNew=true` ; UI shows NEW pill	API	P1
TC-NEW-02	Old model is not NEW	Set a row `first_seen_at = now-8d`	Excluded from `/api/models/new` ; `isNew=false` ; no pill	API	P1
TC-NEW-03	Badge ages out at +7d	Model first seen exactly 7 days + 1s ago	No longer NEW	API	P2
TC-NEW-04	Refresh does not reset firstSeenAt	Note a row's firstSeenAt → Refresh Now → re-check	`first_seen_at` unchanged; badge persists its remaining window	API	P1

G. REST API

ID	Objective	Request	Expected result	Type	Pri
TC-API-01	Paginated list envelope	`GET /api/models`	`{content,totalElements,page,size}` ; page 0, size 50, sort newest	API	P1
TC-API-02	Category filter	`?category=OCR`	Only OCR rows	API	P1
TC-API-03	New-since filter	`?newSince=3`	Only models first seen within 3 days	API	P2
TC-API-04	Sort options	`?sort=downloads` / `likes` / `newest`	Ordered accordingly (e.g. downloads desc)	API	P1
TC-API-05	Case-insensitive search	`?search=QWEN` and `?search=qwen`	Same non-empty result on id/author substring	API	P1
TC-API-06	Pagination & size cap	`?page=1&size=5` ; `?size=9999`	Correct page; size clamped to ≤200	API	P2
TC-API-07	New-this-week endpoint	`GET /api/models/new`	Only firstSeen within 7 days	API	P1
TC-API-08	Categories with counts	`GET /api/categories`	All 8 categories with `{name,label,count}`	API	P2
TC-API-09	Manual sync endpoint	`POST /api/admin/sync`	200 with `{checked,created,updated}` ; a MANUAL `sync_runs` row	API	P1
TC-API-10	Health endpoint	`GET /api/health`	200 `UP` if last success <26h; else 503 `DOWN` ; body has `lastSuccessfulSyncAt`	API	P1

ID	Objective	Request	Expected result	Type	Pri
TC-API-11	JSON shape	Inspect a model object	camelCase keys; `isNew` present; **no** `lastFetchedAt` leaked	API	P2
TC-API-12	CORS	Preflight `OPTIONS /api/models` with `Origin: http://localhost:3000`	`Access-Control-Allow-Origin: http://localhost:3000`, methods GET,POST,OPTIONS	API	P3
TC-API-13	Unknown category param	`?category=BOGUS`	Treated as "all" (no 500)	API	P3

H. Dashboard UI (end-to-end)

ID	Objective	Steps	Expected result	Type	Pri
TC-UI-01	Dashboard renders	Open `http://localhost:8080/`	Header, tabs, stat cards, filter bar, and grid all render	UI	P1
TC-UI-02	Tab per category with NEW counts	Inspect tab bar	"All" + one tab per category; each shows total + green "N new" badge	UI	P1
TC-UI-03	Tab selection filters everything	Click "OCR"	Grid shows only OCR; stat cards update; URL → `?category=OCR`	UI	P1
TC-UI-04	Stat cards accuracy	Per selected tab	Models-in-view, New-this-week, Most-downloaded match the DB for that category	UI	P2
TC-UI-05	Sort control	Choose "Most downloads"	Grid reorders by downloads desc	UI	P2
TC-UI-06	Search (debounced)	Type "qwen"	Grid filters after ~300 ms; URL → `?search=qwen`	UI	P1
TC-UI-07	"New only" checkbox	Tick it	Only NEW models shown; URL → `?new=true`	UI	P2
TC-UI-08	NEW pill correctness	Compare recent vs backdated rows	Pill only on rows first-seen within 7 days	UI	P1
TC-UI-09	Model ID links out	Click a Model ID	Opens `https://huggingface.co/{id}` in a new tab	UI	P2
TC-UI-10	Tooltips on truncated text	Hover each truncated cell (ID, author, category, pipeline tag, description, counts, dates)	Full value shown in a tooltip	UI	P1
TC-UI-11	Refresh Now	Click Refresh now	Progress shows; success notification; grid, tab badges, and stat cards update	UI	P1
TC-UI-12	Shareable deep link	Open `/?category=VLM&sort=downloads&search=x`	Filters applied on load; correct tab selected	UI	P2
TC-UI-13	Empty state	Filter/search with no matches (or empty DB)	Graceful empty grid / "No models yet" message	UI	P3
TC-UI-14	Reads from DB only	Load dashboard while HF is blocked	Dashboard works fully (no live HF call on page load)	UI	P1

I. Scheduler

ID	Objective	Steps	Expected result	Type	Pri
TC-SCH-01	Disabled when no cron	Start with `sync.cron` unset	Cron resolves to `-` (disabled); no scheduled runs	OPS	P1
TC-SCH-02	Daily 8 AM cron	`sync.cron="0 0 8 * * *"` (default now)	Fires once daily at 08:00 local while app runs	OPS	P1
TC-SCH-03	Prod profile enables it	`SPRING_PROFILES_ACTIVE=prod`	8 AM cron active	OPS	P2

ID	Objective	Steps	Expected result	Type	Pri
TC-SCH-04	Env override	`SYNC_CRON="*/30 * * * * *"`	Fires every 30 s (verification only)	OPS	P2
TC-SCH-05	One-time kickoff	`SYNC_KICKOFF="<future ISO>"`	Runs once at that instant (recorded SCHEDULED); past time → skipped + WARN; invalid → WARN	OPS	P2
TC-SCH-06	No overlapping runs (ShedLock)	Trigger concurrent/near-simultaneous runs	Only one executes; a `shedlock` row is held; the other tick is skipped	INT	P1
TC-SCH-07	Scheduled run is audited	After a scheduled fire	`sync_runs` row with `trigger_source=SCHEDULED`	INT	P2
TC-SCH-08	In-JVM requirement	Stop the app before the fire time	The scheduled run does NOT fire (documented: app must be running)	OPS	P2

J. Health & audit trail

ID	Objective	Steps	Expected result	Type	Pri
TC-AUD-01	Every run audited	Trigger a sync	`sync_runs` row transitions RUNNING → SUCCESS with counts	INT	P1
TC-AUD-02	Failures are audited	Force a sync failure	Row recorded as FAILURE with `error_message`; sync itself not silently lost	INT	P2
TC-AUD-03	Health freshness window	Last success 25 h vs 27 h ago	200 UP vs 503 DOWN respectively	API	P1
TC-AUD-04	No success yet	Fresh DB, no sync	Health = 503 DOWN	API	P2

K. Resilience / error handling

ID	Objective	Steps	Expected result	Type	Pri
TC-RES-01	Dashboard independent of HF	Point HF `base-url` at unreachable host; reload	`/`, `/api/models`, `/api/health` all 200 from Postgres; a sync yields `checked=0` + fetch-failed warnings	OPS	P1
TC-RES-02	Rate-limit recovery	Sustained 429s	Retry once then skip; next run recovers those tags	INT	P2
TC-RES-03	DB unavailable	Stop Postgres, hit API	Errors surfaced (5xx); on restart, app reconnects and serves	OPS	P2
TC-RES-04	Unexpected HF fields	HF adds/removes JSON fields	`@JsonIgnoreProperties(ignoreUnknown=true)` → no deserialization crash	INT	P2

L. Configuration

ID	Objective	Steps	Expected result	Type	Pri
TC-CFG-01	Add/remove tracked tag or keyword	Edit `model-radar.tracking` in `application.yml`; restart	The new tag/keyword is queried on the next sync; a removed one is not	OPS	P1
TC-CFG-02	Datasource override	Set `SPRING_DATASOURCE_URL/_USERNAME/_PASSWORD`	App connects to the overridden DB	OPS	P2
TC-CFG-03	README fetch cap	Set `readme-fetch-limit=0`	No README calls; descriptions stay null	OPS	P3
TC-CFG-04	Port collision documented	Native Postgres on 5432 present	Compose maps 5433; app connects; README explains	OPS	P3

M. Non-functional

ID	Objective	Steps	Expected result	Type	Pri
TC-NFR-01	Dashboard latency	Load dashboard	Renders quickly; no synchronous HF calls in the read path	OPS	P2
TC-NFR-02	Large dataset paging	3,000+ rows	Grid stays responsive via lazy 50-row pages (server-side)	UI	P2
TC-NFR-03	Access control	Review security posture	No auth by design (internal tool, network-scoped); documented, not a defect	OPS	P3
TC-NFR-04	Sync duration	Time a full sync	Completes in a bounded time (~30 s incl. up to 300 README fetches)	OPS	P3

Acceptance mapping (Section 10 of the spec)

Spec criterion	Covered by
Automated ≥ once/day	TC-SCH-02, TC-SCH-07, TC-SYNC-07
Only Section 4 categories	TC-CAT-04, TC-CAT-01/02
All Section 5 fields	TC-DB-02, TC-META-* , TC-SYNC-05
NEW badge (last 7 days)	TC-NEW-01..04, TC-UI-08
Filter + sort work	TC-API-02/04/05, TC-UI-03/05/06
Dashboard reads stored data (not HF)	TC-RES-01, TC-UI-14
Runs unattended 7 days	TC-SCH-02/08, TC-DB-03, TC-AUD-03 (monitor via `/api/health`)
README to run/restart	Top-level `README.md` runbook